INDEPENDENT STRATEGY + EXECUTION REVIEW

Apex Quant Funded-100K Qualification Audit

Book C / C_FUNDED_V2

NO FUNDED STRATEGY - DO NOT DEPLOY

1.10%	0.485	3.18%	1.31%
Synthetic eval annualized	Synthetic eval Sharpe	Synthetic eval max DD	Raw-quote peak stop risk

Evidence ceiling: retrospective daily-OHLC replay with unconverted mixed quote currencies. This is not a true blind test, account-currency result, executable venue replay, or guarantee of passing, retaining, or profiting from a funded account.

Decision date: 3 September 2026

Executive decision

Book C at 0.85% remains the incumbent and best-documented return/drawdown compromise among the Apex configurations evaluated in this audit; it is not the leader on every individual metric, and it is not a funded-account strategy. Its historical 14.01% maximum drawdown and -3.68% worst close-to-close day are outside a defensible safety envelope for a hard-loss account, before intraday excursions, gaps, spread, swaps, rejected orders, and currency conversion are added.

The new `C_FUNDED_V2` research design materially reduces nominal exposure and adds stop-aware backtest accounting. A separate fail-closed guard component has been unit-tested, but it is not integrated with the replay, paper engine, or venue state. The candidate has two decisive observed screening failures in the synthetic replay, plus independent data and implementation blockers:

1. **The provisional return diagnostics are too weak.** On the frozen 2023-2024 pseudo-out-of-sample window, the mixed-quote evaluation model produced 1.10% annualized with Sharpe 0.485 and profit factor 1.121. The mixed-quote payout model produced 0.50% annualized with Sharpe 0.342 and profit factor 1.097. Under doubled modeled costs, payout annualized return fell to 0.076% and Sharpe to 0.058.
2. **The designed loss cap is not an invariant.** Within the same unconverted model, carried positions allowed current mark-to-stop loss to rise to 1.310% of its synthetic capital base against a 0.90% evaluation design cap, and to 0.766% against a 0.60% payout design cap. New entries are blocked when the cap is full, but existing positions are not continuously de-risked. A stop being present is therefore not the same as the account being funded-safe.

Exact account-currency conversion could change those percentages, so they are not broker-account statistics. It cannot remove the independent blockers: exact product/contract inputs are unfrozen, ordinary costs are absent from planned-loss reservation, next-open orders are not authoritatively revalidated, total open-plus-pending risk is not atomically reserved, the persistent guard is not replay- or venue-integrated, concentration is unreconciled, and no blind forward evidence exists. No candidate evaluated in this report passes the combined speed, drawdown, execution, and evidence requirements. The honest result is no promotion, no website funded label, and no paid challenge.

What was tested

Two no-retuning protocols were frozen before their respective experiments:

- `engine/data_store/funded_100k_prereg_2026-09-03.md`
- `engine/data_store/funded_100k_v2_prereg_2026-09-03.md`

V1 exposed a units error in the original funded-sizing idea: at a fresh 100,000 account, 0.85% of a 3,000 daily buffer is only 25.50 units of planned risk per trade. A smoke result was intentionally not retained as qualification evidence; the frozen formula itself was enough to reject V1 as economically mis-scaled.

V2 corrected the cash-risk formula. It keeps Book C's frozen `[63, 126, 252]` signal ensemble and managed exits while separating two irreversible modes:

Mode	Designed base risk per instrument	Designed aggregate stop-risk cap	Designed gross cap	Max pos itions	Intended day block / flatten	Intended cycle halt
Evaluation	0.35% of `min(equity, initial)`	0.90%	0.60x	5	0.90% / 1.50%	5.00%
Payout	0.25% of `min(equity, initial)`	0.60%	0.45x	4	0.60% / 1.20%	4.00%

These are frozen research design limits, not controls proven active on a venue. Every synthetic entry carries a protective stop in the backtester. The replay gives stops priority on an ambiguous daily bar, applies a worse opening fill when price gaps through the stop, charges commissions per fill, and checks entry-bar stops. These are conservative corrections, not evidence that an actual venue accepted or executed protection at the requested price.

Provisional 2023-2024 mixed-quote diagnostic

The result below is the hardened validation-only replay in engine/data_store/validation/funded_100k_v2_gate_2026-09-03.json. Static and completed-EOD-trailing variants were identical in this window. The artifact explicitly labels its currency basis UNCONVERTED_RAW_QUOTE_CURRENCY: balances, P&L, exposure, and stop risk combine different quote currencies without executable account-FX conversion. The percentages below are therefore **synthetic within-model diagnostics**, not account-currency returns, drawdowns, or cap measurements.

Synthetic cell	Total return	Annualized	Sharpe	Profit factor	Max DD	Worst daily OHLC bound	Mean month	95% lower mean-month bound
Evaluation, base costs	+2.217%	1.101%	0.485	1.121	3.176%	-0.803%	+0.096%	-0.295%
Evaluation, doubled costs	+2.225%	1.105%	0.490	1.125	3.216%	-	-	-
Payout, base costs	+0.997%	0.497%	0.342	1.097	2.424%	-0.546%	+0.043%	-0.217%
Payout, doubled costs	+0.152%	0.076%	0.058	1.020	3.699%	-	-	-

The small improvement in the evaluation doubled-cost cell is a path/selection effect: costs alter which constrained candidates fit, so it must not be interpreted as evidence that higher costs improve the strategy. The payout cell shows fragility inside this model. None of the figures is an accurate expected funded-account return until contract values and causal FX conversion are applied.

The corrected run is explicitly non-binding because deep diagnostics were stopped once deterministic screening gates had failed. Completing hundreds of order permutations or 100,000 whole-day bootstrap paths cannot repair those observed values inside the same synthetic model. Exact currency/contract replay may change the values, but it is itself a mandatory data gate rather than permission to promote the strategy. The runner prevents reduced diagnostic arguments from crossing the qualification boundary.

Gate decision

Required property	Threshold	Observed	Decision
Evaluation Sharpe	>=0.75	0.485, mixed-quote	Provisional screen fail
Evaluation profit factor	>=1.15	1.121, mixed-quote	Provisional screen fail
Payout annualized return	>=4.00%	0.497%, mixed-quote	Provisional screen fail
Payout doubled-cost annualized return	>=2.00%	0.076%, mixed-quote	Provisional screen fail
Payout monthly mean, lower 95% bound	>0	-0.217%, mixed-quote	Provisional screen fail
Evaluation aggregate stop risk	<=0.90% of capital base	1.310%, raw-quote sum	Provisional screen fail
Payout aggregate stop risk	<=0.60% of capital base	0.766%, raw-quote sum	Provisional screen fail
Exact contract, account-FX, bid-ask, and stress replay	Required	Missing / not integrated	Data-blocked
Planned loss includes entry/exit costs	Required	Stop distance only	Data-blocked
Atomic open-plus-pending reservation and next-open revalidation	Required	Not integrated	Data-blocked
Venue stop acknowledgement and persistent guard transitions	Required	No event stream / no integration	Data-blocked
Official opened-position trading-day ledger	Required	Backtester opening count only	Data-blocked
Reconciled terminal positions and concentration	Required	Incomplete	Data-blocked
True blind forward evidence	>=6 months and >=100 completed trades	Not available	Fail

Whole-day bootstrap output is now marked diagnostic-only. It cannot be a binding survival probability because it does not rebuild positions and stops at every sampled event, enforce block/flatten/cycle-halt transitions, or reproduce pending-order reservations. DSR is also data-blocked: the historical trial ledger does not record the annualization convention for every prior Sharpe observation. Concentration is data-blocked until terminal open positions and partial realizations reconcile exactly to final equity.

Why a stop loss does not by itself protect a funded account

A protective stop bounds loss only under its execution assumptions. It does not guarantee the requested fill through an overnight gap, a disconnected terminal, rejected/cancelled protection, stale FX, partial liquidation, or insufficient liquidity. It also does not automatically cap the **sum** of all open and pending risks.

The corrected backtest/risk boundary now rejects missing, non-finite, zero, or wrong-side stops and targets. It also rejects non-finite price, volatility, ATR, FX, sizing, cap, unit, notional, and risk values. This closes a real fail-open defect where a `NaN` stop could previously create a permitted position whose exit comparison never fired.

The isolated guard component correctly fails closed in unit tests on stale, corrupt, or out-of-order state and persists daily/cycle latches. It is not yet called by the V2 replay, website paper engine, or a venue event loop. Two production properties also remain unimplemented:

- carried positions are not deterministically trimmed or re-stopped whenever total mark-to-stop risk rises above the cap; and
- approvals do not create one atomic broker-backed reservation for the complete open-plus-pending book, so independent approvals cannot be treated as separately spendable budgets.

Until those are solved with authoritative venue state, a trade cannot honestly be described as impossible to lose 10,000. The proposed internal flatten thresholds create distance from the firm's hard floor, but a sufficiently large gap or execution failure can jump both a stop and a software threshold.

Existing-engine comparison

These figures come from overlapping studies with different windows, universes, and evidence quality. They compare the evaluated candidates; they do not prove an exhaustive or universally best ranking.

Candidate	Return evidence	Sharpe	Drawdown/tail evidence	Funded assessment
Book C 0.85%	12.87% CAGR; 1,767 account units/month	1.144	14.01% DD; -3.68% worst close day	Incumbent documented compromise; not funded-safe
Book C 0.75%	11.56% CAGR; 1,503 units/month	1.149	12.80% DD	Safer, still outside hard-loss envelope
Book H 0.50%	4.95% annualized; 413 units/month	0.922	8.2% forward p95 DD	Better tail, too slow and not execution-certified
Book R-252	17.63% CAGR	0.972	23.63% DD; -6.54% day	Reject
Book R 3ATR sensitivity	6.84% CAGR	0.987	9.01% DD; -2.98% day	Post-result sensitivity; not promotable
Book N base	9.37% CAGR	0.936	12.13% DD	Current-universe selection bias; reject
C_FUNDED_V2 payout	0.50% synthetic annualized	0.342	2.42% synthetic DD; raw-quote cap overrun	Provisional screen fail and operationally blocked

Book C 0.75% previously achieved only an optimistic close-only 54.8% all-start 2-Step pass rate, with a 264-day median among passers. A different 0.75%/2.5%-cap frontier cell had 4.6% CAGR and only 13.1% of resampled paths reached +10% inside 252 sessions. These are overlapping retrospective proxies, not independent pass probabilities. They demonstrate the speed/safety conflict rather than solve it.

Best product shape, if the engineering and evidence later pass

For Book C's roughly 21-session holding logic, the closest current FTMO product fit is **2-Step Swing**, not 1-Step Standard. FTMO currently states that 2-Step uses +10% Challenge and +5% Verification targets, a 5% maximum daily loss, a static 10% maximum loss, and at least four trading days on each of which a position is opened. The profit target is met only with all positions closed. FTMO also states that Swing is exclusive to 2-Step and permits overnight, weekend, and news holding. Standard funded-stage restrictions would force closures and therefore change Book C's tested strategy. See FTMO Trading Objectives, FTMO Swing account type, and overnight/weekend rules.

This is a product-fit conclusion, not a recommendation to buy a challenge. FTMO requires continuous compliance using live equity including floating P/L, swaps, and commissions, and its exact symbols, spreads, specifications, and trading hours are platform-specific. Algorithmic trading is allowed only when legitimate, properly risk-managed, and replicable under real-market conditions. See FTMO strategy rules and FTMO symbols/specifications.

C_FUNDED_V2 was **not** replayed as an exact FTMO 2-Step Swing account. It used a synthetic 3% internal daily envelope and evaluated both static and completed-EOD-trailing maximum-loss variants. The product section identifies a plausible future container only; exact FTMO qualification requires the selected platform's rules, opened-position trading days, contract values, account currency, and event-level fills.

Better-strategy research direction

No Apex candidate evaluated in this audit qualifies. The next defensible family is a new, platform-native diversified time-series trend strategy across the exact available equity indices, major FX, metals/commodities, and-where the selected platform actually offers them-rates/bond markets. It should not reuse today's selected-stock-heavy universe or substitute convenient ETFs for unavailable contracts.

The hypothesis has a sound research basis: Moskowitz, Ooi, and Pedersen document time-series momentum across liquid equity-index, currency, commodity, and bond futures (paper); Moreira and Muir report that reducing exposure when volatility is high improved factor Sharpe ratios in their historical samples (NBER). Neither result guarantees a profitable implementation. Daniel and Moskowitz show that momentum can suffer persistent crashes, especially after market declines, during high volatility, and into rebounds (NBER paper). Those crash states must be designed into the stress protocol, not cited away.

Implementation plan from here

1. **Freeze the actual product.** Record firm, product, account currency, platform, timezone/reset, automation permission, news/weekend rules, maximum allocation, and exact symbol list. No generic "100k funded" profile is sufficient.
2. **Acquire executable data.** Export contract multipliers, tick/lot sizes, leverage/margin, commissions, swaps, and at least one-minute bid/ask plus order/fill/stop-acknowledgement history. Add timestamped account-currency FX conversion.
3. **Build the execution safety invariant.** Revalidate every pending order from authoritative opening equity; atomically reserve all open-plus-pending planned loss; include ordinary costs; and trim/cancel/flatten whenever total stressed risk exceeds its cap. Confirm fills and persist latches across restarts.
4. **Pre-register a new candidate.** Either fix Book C as `C_FUNDED_V3` without using V2 outcomes to optimize it, or freeze the platform-native diversified trend family above. Define the universe, signals, stops, costs, phase rules, and every pass gate before viewing results.
5. **Run event-level qualification.** Fresh accounts for build/interim/validation; static and any applicable trailing floors; Challenge, Verification, and payout modes; exact four-day accounting; gap, spread, swap, outage, missed-stop, and partial-fill stresses; fixed-model combinatorial OOS; concentration reconciliation; and a unit-complete DSR reference.
6. **Shadow only after a full pass.** Run a separate 100,000-unit paper account unchanged for at least six months and 100 completed trades. Require zero official breaches, zero unexplained backtest/paper parity differences, and venue-stop acknowledgement on every entry.
7. **Paid use requires a second decision.** A retrospective or shadow pass is not automatic authority to buy a challenge or route orders. Recheck the then-current rulebook and run a final go/no-go audit.

The qualitative/LLM validation node should remain shadow-only as well. Point-in-time fundamentals, filings, insider data, and short-interest history are not available across the full historical universe, so it cannot yet be backtested causally as a hard override. It may log an independent score, but it must not rescue or veto `C_FUNDED_V2` until it passes a separately frozen walk-forward gate.

Engineering changes and verification

The isolated implementation adds:

- stop-aware funded equity traces with conservative gap-through-stop fills;
- entry-bar stop enforcement and per-fill commissions;
- correct cash-risk bases and separate evaluation/payout limits;
- an isolated, account-scoped, DST-aware, fail-closed persistent guard component, not yet execution-integrated;

- scalar/vector rule replay requiring balance, equity, and verified-flat target state, with explicit opened-position counts instead of a closed-P&L trading-day proxy;
- hard rejection of non-finite or wrong-side risk inputs;
- immutable source/config/data/prior-trial manifests, no-retry ledgers, distinct output paths, and a qualification boundary that reduced runs cannot cross; and
- explicit machine-readable blockers for every missing funded-critical property.

All targeted funded/risk tests and the complete repository test suite passed locally after the corrections. Python compilation and `git diff --check` also passed. This is local verification, not an external CI or venue certification. No Book A/B/C/R paper state, website strategy, broker position, paid challenge, or external account was changed by the qualification run.

Limitations

- The 2023-2024 window is pseudo-OOS relative to the latest design, not genuinely unseen history; earlier Book C research has already inspected it.
- Daily OHLC supplies a conservative set of co-extreme bounds, not the true event order inside a bar.
- Historical P&L and exposure mix quote currencies without executable account-currency conversion.
- The data use today's selected instruments historically; delisted names and point-in-time membership are incomplete.
- Stops reduce ordinary losses but cannot guarantee fills through gaps or venue failures.
- Backtests and resampling do not guarantee future returns, passing a challenge, retaining an account, or receiving payouts.

Claim-to-source ledger

Claim supported	Source	Use and boundary
Book C 0.85% statistics, concentration, funded proxy, no true blind interval	engine/data_store/BOOK_C_DEEP_AUDIT_2026-08-19.md; engine/data_store/validation/book_c_risk_frontier_2026-08-20.json	Local reproducible research; daily data and current-universe caveats apply
V1 sizing-units rejection	engine/data_store/funded_100k_prereg_2026-09-03.md; engine/scripts/run_funded_100k_gate.py	Formula-level rejection; no V1 smoke result is presented as evidence
V2 synthetic metrics, raw-quote cap overrun, blockers, source/data hashes	engine/data_store/validation/funded_100k_v2_gate_2026-09-03.json; dedicated V2 ledger	Non-binding, unconverted 2023-2024 diagnostic; not account-currency or venue-certified
Earlier funded frontier speed/safety trade-off	engine/data_store/prop_condition_frontier.md	Retrospective EOD Monte Carlo; not independent pass odds
Book H 0.50% comparison statistics	engine/data_store/profit_frontier_2026-07-23.md	Earlier local frontier study; different scope from V2
Book R stop-overlay rejection	engine/data_store/validation/book_r_stop_overlay_audit_2026-09-03.md	Retrospective causal test; sensitivity not promotable
Book N selection-bias warning	engine/data_store/validation/book_n_qe_10y_causal_audit_2026-08-28.md	Current static universe; no delisted-member reconstruction
Risk boundary, opened-position day accounting, and local regression verification	engine/tests/test_risk.py; engine/tests/test_funded_guard.py; engine/tests/test_funded_simulator.py; engine/tests/test_portfolio_funded_trace.py	Executable local tests; not broker integration or external CI evidence
Current FTMO targets, daily/max-loss mechanics, four-day minimum	FTMO Trading Objectives	Official current rule summary; must be rechecked before any purchase
Swing availability and holding permissions	FTMO Swing FAQ; holding FAQ	Official current product-fit evidence
Algorithmic strategy permissibility and replicability requirement	FTMO strategy FAQ	Official policy; does not certify this engine
Platform-specific symbols/specifications	FTMO symbols	Exact values must come from the selected platform/export
Diversified time-series momentum hypothesis	Moskowitz, Ooi & Pedersen	Academic historical evidence; hypothesis, not forecast
Volatility-management hypothesis	Moreira & Muir	Academic factor evidence; implementation-specific testing required
Momentum crash risk	Daniel & Moskowitz	Academic tail-risk evidence motivating stress tests

Final verdict

Book C remains the incumbent, best-documented compromise among the Apex configurations evaluated here for unconstrained paper research. It is not a funded-account strategy, and no evaluated candidate has passed the funded qualification gate. C_FUNDED_V2 appears lower-risk inside a provisional mixed-quote model, but that model screens poorly and is operationally/data-blocked. Keep it research-only, do not show it as funded-ready, and build the next candidate only after the exact funded product and executable data are frozen.